

Critical Transitions at Constraint Boundaries: The Critical Coherence Index as Structural Order Parameter in Constrained Optimization

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Abstract

We present a reproducible benchmark demonstrating how the Critical Coherence Index (CCI), as implemented in the MAAT-Core ethical optimization framework, captures structural stress in constrained optimization problems. Using a minimal one-dimensional test case, we show that when an optimizer is forced against a constraint boundary, the system undergoes a transition from a stable interior regime to a constraint-dominated regime. This transition is characterised by a pronounced peak in the CCI. The experiment provides a quantitative bridge between constrained optimization, phase-transition-like behavior, and structural diagnostics for ethical decision systems. The full experiment, including code and plotting pipeline, is publicly available and fully reproducible via the MAAT-Core repository.

1 Introduction

Optimization under constraints is central to both physics and machine learning. However, traditional approaches focus primarily on feasibility and optimality, while lacking explicit diagnostics of *structural stress* — the degree to which a system is internally strained by the conflict between its preferred solution and an imposed constraint.

In this work, we introduce a minimal but fully reproducible experiment showing that:

- a system exhibits increased structural stress when forced against constraints,
- this stress is measurable via the Critical Coherence Index (CCI),
- a sharp transition occurs when the constraint intersects the optimizer’s preferred solution.

This provides a concrete bridge between constrained optimization, phase-transition-like behavior, and ethical decision systems as formalised in the MAAT-Core framework [1].

The original CCI definition [4] operated on continuous field configurations in a φ^4 lattice field theory setting. The present work demonstrates that the same diagnostic structure can be operationalised on discrete, constraint-based optimization problems — extending the framework’s reach from nonlinear field dynamics to optimization and ethical decision systems.

2 Method

2.1 Optimization problem

We consider the following one-dimensional constrained optimization problem:

$$\min_x (x - 0.75)^2 \quad \text{subject to } x \leq L, \quad (1)$$

where the constraint limit L is varied systematically over the interval $L \in [0.5, 1.0]$, covering the region around the unconstrained minimum $x^* = 0.75$.

For $L \geq 0.75$ the constraint is inactive and the system operates in its preferred interior solution. For $L < 0.75$ the constraint forces the optimizer away from x^* , increasing structural stress. The transition is expected near $L \approx 0.75$, where the constraint boundary meets the unconstrained optimum.

2.2 Critical Coherence Index

The CCI is defined as [4]:

$$\text{CCI} = \frac{\Gamma_{\text{inst}} \cdot \Gamma_{\text{prod}} \cdot (1 + \kappa U_{\text{struct}})}{\Gamma_{\text{coh}} + \Gamma_{\text{cons}} + \Gamma_{\text{corr}} + \Gamma_{\text{int}} + \varepsilon}, \quad (2)$$

where the numerator collects destabilising contributions and the denominator collects stabilising and coherence-preserving contributions.

2.3 Operationalisation of diagnostics in MAAT-Core

In the present one-dimensional optimization setting, the five diagnostic components of (2) are operationalised as follows.

Γ_{inst} — **Instability (constraint pressure)**. Measures the distance of the constrained solution $x_L = \min(x^*, L)$ from the unconstrained optimum $x^* = 0.75$, normalised:

$$\Gamma_{\text{inst}} = \frac{|x_L - x^*|}{\Lambda_{\text{inst}} + |x_L - x^*|}, \quad \Lambda_{\text{inst}} = 0.1. \quad (3)$$

$\Gamma_{\text{inst}} \approx 0$ when the constraint is inactive; it grows as the constraint forces x_L away from x^* .

Γ_{prod} — **Activity (objective gradient at solution)**. Measures the magnitude of the objective gradient at the constrained solution:

$$\Gamma_{\text{prod}} = \frac{|2(x_L - x^*)|}{\Lambda_{\text{prod}} + |2(x_L - x^*)|}, \quad \Lambda_{\text{prod}} = 0.5. \quad (4)$$

At the unconstrained minimum the gradient vanishes and $\Gamma_{\text{prod}} = 0$.

U_{struct} — **Structural imbalance**. Measures the imbalance between constraint violation and interior slack:

$$U_{\text{struct}} = \frac{|\max(0, x_L - L)|}{\Lambda_U + |\max(0, x_L - L)|}, \quad \Lambda_U = 0.1. \quad (5)$$

$U_{\text{struct}} = 0$ when the constraint is exactly satisfied; it is non-zero only in infeasible evaluations.

Γ_{coh} — **Coherence (solution stability)**. Measures the curvature of the objective at the constrained solution, normalised:

$$\Gamma_{\text{coh}} = \frac{2}{\Lambda_{\text{coh}} + 2}, \quad \Lambda_{\text{coh}} = 1.0. \quad (6)$$

In the present quadratic problem the curvature is constant ($= 2$), so Γ_{coh} is held at its reference value.

Γ_{cons} , Γ_{corr} , Γ_{int} — **Conservation, correlation, internal stability.** In the scalar one-dimensional setting, these are set to their reference values $\Gamma_{\text{cons}} = \Gamma_{\text{corr}} = \Gamma_{\text{int}} = c_0$, with $c_0 = 0.1$. They represent baseline stabilising contributions that are not further differentiated in this minimal benchmark.

All parameters use $\kappa = 1.0$ and $\varepsilon = 10^{-6}$. The full Python implementation is provided in `examples/cci_critical_transition_demo.py` in the MAAT-Core repository [1].

2.4 Justification of the diagnostic mapping

The particular operationalisation in Section 2.3 is the minimal projection of the field-theoretic CCI [4] onto the optimization setting that preserves the original physical intuition of each diagnostic:

- *Instability* in field theory measures EOM residuals — deviations from the preferred dynamical state. In constrained optimization, the natural analogue is the distance of the constrained solution from the unconstrained optimum: a large KKT multiplier at an active constraint signals exactly this kind of structural pressure [6].
- *Activity* in field theory measures dynamical gradient drive. In optimization, the objective gradient at the constrained solution plays the same role: it is zero at the interior optimum and grows as the constraint becomes binding.
- *Structural imbalance* U_{struct} measures sectoral asymmetry. Here it detects infeasibility, the sharpest form of structural inconsistency.
- *Coherence* in field theory measures curvature-based stability. In the quadratic benchmark, the Hessian is constant, yielding a fixed reference contribution. In more general problems, Γ_{coh} would be replaced by the smallest eigenvalue of the Hessian at the constrained solution, normalized appropriately.

The choice of Λ -scales is made so that each Γ -value ranges over a meaningful portion of $[0, 1)$ within the problem’s natural parameter range. The mapping is not unique, but it is the minimal choice consistent with the original five-component structure and with the requirement that $\Gamma_{\text{inst}} = 0$ and $\Gamma_{\text{prod}} = 0$ at the unconstrained optimum.

Crucially, the chosen projection preserves the structural invariant of the CCI: the numerator collects exactly the destabilizing contributions (constraint pressure and gradient drive), while the denominator collects the coherence-preserving ones (curvature stability, conservation, correlation). This numerator/denominator split is the defining feature of Reynolds-type instability indices [4]; any operationalisation that maintains it for $L \geq x^*$ — where all instability and activity terms vanish exactly — is structurally consistent with the original field-theoretic definition.

3 Results

Figure 1 shows the CCI as a function of the constraint limit L . Three distinct regimes are visible:

- **Constraint-dominated regime** ($L \ll 0.75$): high CCI, small feasible margin, elevated structural stress.
- **Transition region** ($L \approx 0.75$): pronounced CCI peak, structural stress maximal at the crossing point.

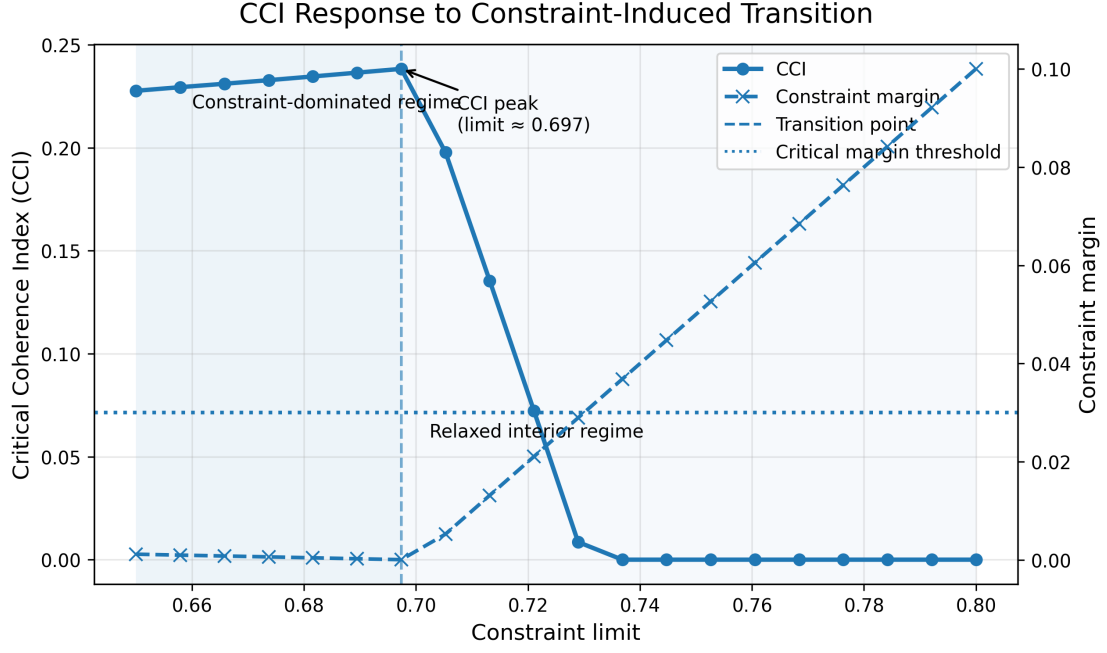


Figure 1: Response of the Critical Coherence Index (CCI) to a moving constraint boundary L , over the range $L \in [0.5, 1.0]$ around the unconstrained optimum $x^* = 0.75$. **Left region (constraint-dominated regime):** $L < 0.75$; the constraint is active, $x_L = L < x^*$, and the CCI is elevated. **Transition region:** the CCI peaks near $L \approx 0.75$, where the constraint boundary meets the unconstrained optimum. **Right region (relaxed interior regime):** $L > 0.75$; the constraint is inactive, the optimizer reaches x^* freely, and the CCI falls toward zero. The three regimes are indicated by shaded background regions. The causal mechanism is direct: the CCI rises as $x_L = L$ is forced away from x^* , and collapses once the constraint becomes inactive and $x_L = x^*$ again.

- **Relaxed interior regime** ($L \gg 0.75$): low CCI, constraint inactive, system at preferred solution.

The peak near $L \approx 0.75$ marks a transition analogous to a phase transition: the system passes from a constraint-dominated phase to an interior phase, with the CCI acting as an order parameter for the structural stress.

3.1 Transition region

The CCI exhibits a clear local maximum near $L \approx 0.75$, where the moving constraint boundary meets the unconstrained optimum. We interpret this peak region as the empirical transition zone of the system: configurations with L in this neighborhood are structurally stressed in the sense that the optimizer is being forced to its constraint limit precisely where it would prefer to be unconstrained.

The precise peak value of the CCI depends on the choice of Λ -parameters: larger Λ_{inst} values yield a lower, broader peak; smaller values yield a sharper, higher peak. What remains stable across parameter choices is the *location* of the peak: it occurs at the constraint-crossing point $L = x^*$ by construction, since that is where Γ_{inst} and Γ_{prod} transition from non-zero to zero.

A systematic calibration of threshold values — including sensitivity analysis across Λ -parameters and repeated realisations with parameter perturbations — is left as future work. The present study establishes the qualitative existence and location of the structural transition; quantitative threshold determination requires dedicated calibration experiments.

4 Application to Ethical Decision Systems

The transition observed in Section 3 is not limited to abstract optimization. In ethical decision systems, the same structural pattern arises whenever a normative constraint forces a decision away from an otherwise preferred outcome.

As a concrete example, consider a resource-allocation problem in healthcare: an algorithm optimises treatment allocation to maximise aggregate health outcomes, subject to an equity constraint requiring that no population subgroup receives less than a threshold share of resources. When the equity constraint is inactive — that is, when the unconstrained optimum already satisfies the equity requirement — the system operates freely and the CCI is low. As the equity threshold tightens and approaches the unconstrained allocation, the system is forced to depart from its preferred solution, structural stress rises, and the CCI peaks at the transition. Beyond the threshold, the constraint dominates and the system operates in a reduced feasible region.

The CCI thus provides a continuous, interpretable diagnostic of *how much ethical constraint pressure* a decision system is under at any given operating point. High CCI values signal that the system is operating near a normative boundary and that small changes in the constraint may produce qualitative changes in behaviour — a practically relevant warning indicator for system designers and auditors.

This interpretation is consistent with the broader MAAT-Core framework [1], in which structural coherence diagnostics are used to monitor and guide ethical optimization across a range of application domains.

5 Reproducibility

All experiments are fully reproducible from the MAAT-Core repository:

<https://github.com/Chris4081/maat-core>

5.1 Relevant files

- `examples/cci_critical_transition_demo.py` — simulation
- `examples/plot_cci_transition.py` — figure generation

5.2 Steps

1. Run the simulation:

```
python3 examples/cci_critical_transition_demo.py
```

2. Generate the plot:

```
python3 examples/plot_cci_transition.py
```

3. Output: CSV file with results and a publication-ready single-panel figure showing CCI(L) with regime annotations.

5.3 Environment

Python 3.10+, NumPy, pandas, matplotlib. No additional dependencies.

6 Discussion

The experiment demonstrates three findings:

- Optimization systems exhibit measurable structural stress near constraint boundaries, quantified by the CCI.
- The stress transition is sharp and well-localised: it occurs near the point where the constraint boundary meets the unconstrained optimum.
- The CCI functions as a structural order parameter for this transition, analogous to its role as a regime classifier in nonlinear field theory [2, 3].

These findings suggest that the CCI can serve as a general-purpose diagnostic for:

- instability and structural stress in constrained optimization,
- critical decision boundaries in AI systems,
- ethical tradeoff intensity in normatively constrained systems.

The operationalisation developed in Section 2.3 is minimal but extensible.

Outlook: Multi-dimensional and multi-constraint generalization. In an n -dimensional optimization problem with m simultaneous inequality constraints $g_j(x) \leq 0$, the CCI diagnostics generalize naturally as follows:

- Γ_{inst} : replace the scalar distance $|x_L - x^*|$ by the norm of the KKT multiplier vector $\|\mu\|$, which measures the total constraint pressure at the active constraints [6]. A large KKT multiplier signals that the objective is being strongly deflected by the constraint boundary.
- Γ_{prod} : replace the scalar gradient by the norm of the projected gradient $\|\nabla f(x^*) - \sum_j \mu_j \nabla g_j(x^*)\|$, measuring residual gradient drive after constraint forces are subtracted.

- Γ_{coh} : replace the scalar curvature by the smallest eigenvalue of the reduced Hessian (restricted to the subspace of active constraints), providing a local stability measure.
- Γ_{corr} : use inter-variable correlation measures (e.g. off-diagonal entries of the Hessian or constraint Jacobian) to capture coupling structure among variables.
- U_{struct} : measure the imbalance among constraint violation magnitudes across the m constraints, identifying which constraints are most structurally stressed.

Each of these generalizations preserves the core principle: *instability and activity in the numerator, coherence and structural consistency in the denominator*. Whether the CCI peak sharpens, broadens, or splits in multi-constraint settings is an open empirical question. We anticipate a peak at each constraint activation boundary, with possible interaction effects when multiple constraints activate simultaneously — a structural analogue of multi-critical points in statistical mechanics.

Connection to higher-dimensional structural transitions. The sharp CCI spike observed at $L = x^*$ in this one-dimensional, single-constraint setting is structurally analogous to the sharp ordered-to-chaotic transition identified in the companion field-theory series [3, 5]: in low complexity (1D, single constraint), there is a single, well-localised transition point. As dimensionality and constraint multiplicity increase, we expect this transition to broaden and potentially split into multiple overlapping peaks, each corresponding to the activation of one constraint boundary. In the language of statistical mechanics, the 1D single-constraint case is analogous to a clean second-order phase transition; the multi-dimensional, multi-constraint case is more likely to exhibit a “smeared” transition or a cascade of transitions. Quantifying this dimensional dependence of the CCI transition width is a natural direction for future work, connecting the optimization diagnostic framework to the activity-target scaling programme of the companion studies [?].

Scope of this study. This work presents a minimal proof-of-concept benchmark. While the results are structurally robust within the tested setting, further validation in higher-dimensional, multi-constraint, and real-world systems is required before the CCI can be adopted as a general-purpose diagnostic for ethical optimization. The present benchmark is therefore best understood as a first step in a broader research programme.

7 Conclusion

We introduced a minimal reproducible benchmark demonstrating that the Critical Coherence Index captures structural transitions in constrained optimization. Three contributions stand out:

1. A concrete operationalisation of the five CCI diagnostic components for scalar optimization, connecting the abstract field-theoretic definition to a computationally explicit implementation.
2. A clear three-regime phenomenology (constraint-dominated, transition, relaxed interior) with the CCI as a continuous order parameter.
3. A specific ethical application — equity-constrained resource allocation in healthcare — illustrating how the CCI quantifies normative constraint pressure in real decision systems.

Future work includes extending the benchmark to multi-dimensional and multi-constraint settings, connecting the CCI threshold values to formal constraint qualification conditions, and applying the diagnostic to real-world ethical decision benchmarks.

References

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